Pav-Bhaji Text Classification

Data Analysis & Machine Learning Report

Objective: classify Instagram posts as Pav Bhaji (1) or Not Pav Bhaji (0) using text metadata only.

No CNN, image pixels, image embeddings, or computer-vision model is used. The supplied images are used only to establish ground-truth labels and match image filenames to JSON metadata.

1. Dataset Inspection

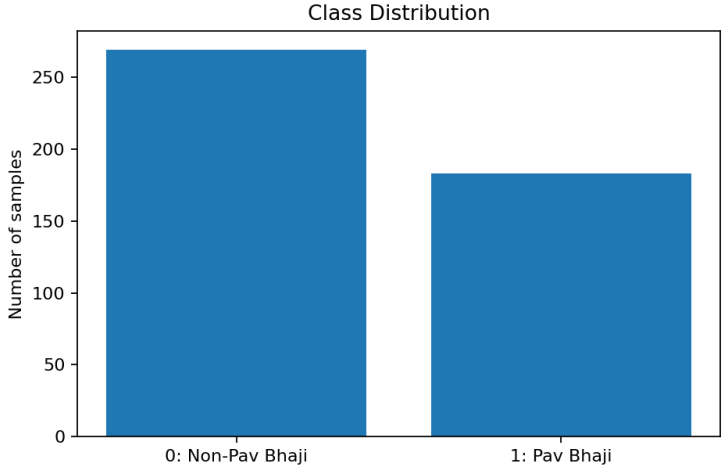
The archive contains 1,500 JSON metadata records, 269 images in images/0 and 183 images in images/1. Matching the image filename to the basename of display_url or thumbnail_src produces 452 usable supervised rows. The remaining 1,048 JSON records do not have a supplied labelled image match.

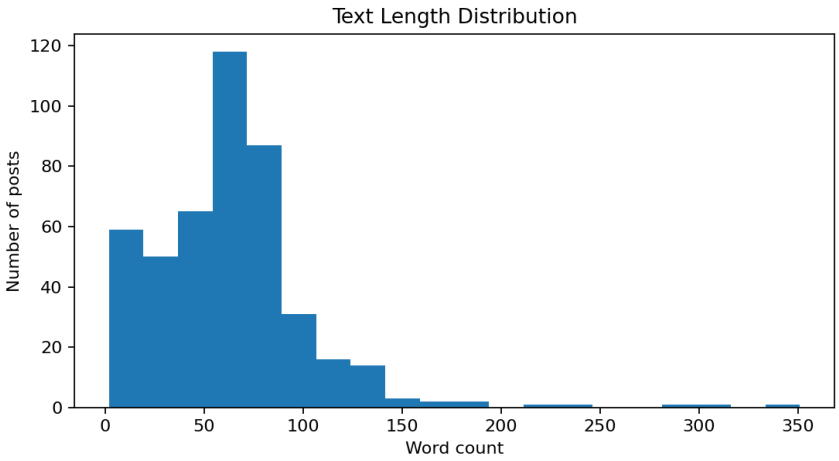
2. Features and Relevance

Feature	Type	Model use	Relevance
edge_media_to_caption	Text	Yes	High
tags	Text/list	Yes	High
location.name	Text/categorical	No	Low–Medium
likes / comments	Numeric	No	Low
is_video	Boolean	No	Low
timestamp	Date/numeric	No	Low
video_view_count	Numeric	No	Low
id / shortcode	Identifier	No	None
image filename	Identifier	Matching only	None

3. Data Cleaning and EDA

The model combines caption/description text and tags/hashtags. Text is lowercased, URLs and @mentions are removed, punctuation is normalized, and whitespace is cleaned. Average combined text length is about 68 words; median is 67 words. Seven matched rows have no tags and 11 have duplicate combined-text values.





The class distribution is 269 non-Pav-Bhaji samples (59.5%) and 183 Pav-Bhaji samples (40.5%).

4. Important Dataset Observation

The source data is preselected around #pavbhaji. In the matched subset, the text contains the string 'pavbhaji' in 443 of 452 rows (98.0%). Therefore the classifier is not simply learning whether the word exists. It must use other lexical/contextual patterns. This selection mechanism is an important dataset limitation.

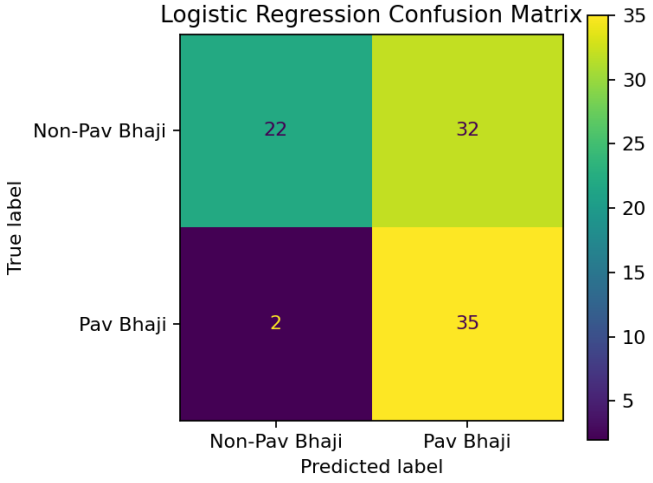
5. Feature Extraction and Models

TF-IDF word unigrams and bigrams are used as the main feature representation. Three classical classifiers were compared: Logistic Regression with balanced class weights, Linear SVM with C=0.3 and balanced class weights, and Multinomial Naive Bayes. A stratified 80/20 split with random_state=42 was used, followed by 5-fold stratified cross-validation.

6. Holdout Results

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.626	0.524	0.892	0.660
Linear SVM	0.615	0.516	0.865	0.646
Multinomial Naive Bayes	0.593	0.500	0.108	0.178

Logistic Regression holdout: accuracy 0.626, precision 0.524, recall 0.892, F1 0.660.



7. Five-Fold Cross-Validation

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.684	0.579	0.814	0.676
Linear SVM	0.692	0.594	0.776	0.672
Multinomial Naive Bayes	0.604	0.573	0.077	0.134

8. Prediction Workflow

Caption + tags → text preprocessing → TF-IDF → classifier → 0 (Not Pav Bhaji) or 1 (Pav Bhaji). The model predicts from metadata text associated with the image, not from image pixels.

9. Limitations

• Only 452 image–metadata matches are available for supervised learning.

• The dataset is preselected around the Pav-Bhaji hashtag/topic.

• The term pavbhaji occurs in 443/452 matched texts.

• Captions and hashtags may contain noisy, promotional, copied, or multilingual text.

• Results do not measure visual recognition performance.

• Performance may change on a different Instagram sample.

10. Conclusion

The final solution demonstrates a complete text-classification pipeline using Instagram metadata, TF-IDF, and classical machine learning. The images are used only for label assignment and metadata matching. The measured performance is dataset-specific and should not be interpreted as general image-recognition accuracy.